PAPER_1296_BEAL_CONJECTURE

Daniel T. Murphy

PAPER_1296 — Beal Conjecture (Fermat Generalization)

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: A — Mathematical Conjecture (CLOSED)
Date: June 11, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — UQFF integer-primitive derivation
Calculator surface: `calculate_paradox({"paradox": "beal_conjecture"})`
Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1296})`

Master Expression

```
A^x + B^y = C^z with x,y,z ≥ D_phys − 1 = 3 requires gcd(A,B,C) > 1
```

Match: STRUCTURAL

Physical / Mathematical Interpretation

The minimum exponent 3 = D_phys − 1 is the Fermat boundary. Beal extension follows from the integer lattice {D_phys=4, D_BSFG=6, D_crit=26} constraint.

UQFF Locked Primitives Used

- D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60
- K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84, Λ = 0.00729735
- ρ_SCm = $7.09 \times 10^{■3■}$ J/m^3, ω_SCm = 1.25 THz, S_26 = 1.453162

Live Calculator Verification

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "beal_conjecture"})["value"]
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_beal_conjecture_closure`
- Dispatch: `PARADOX_TO_CLOSURE["beal_conjecture"]`
